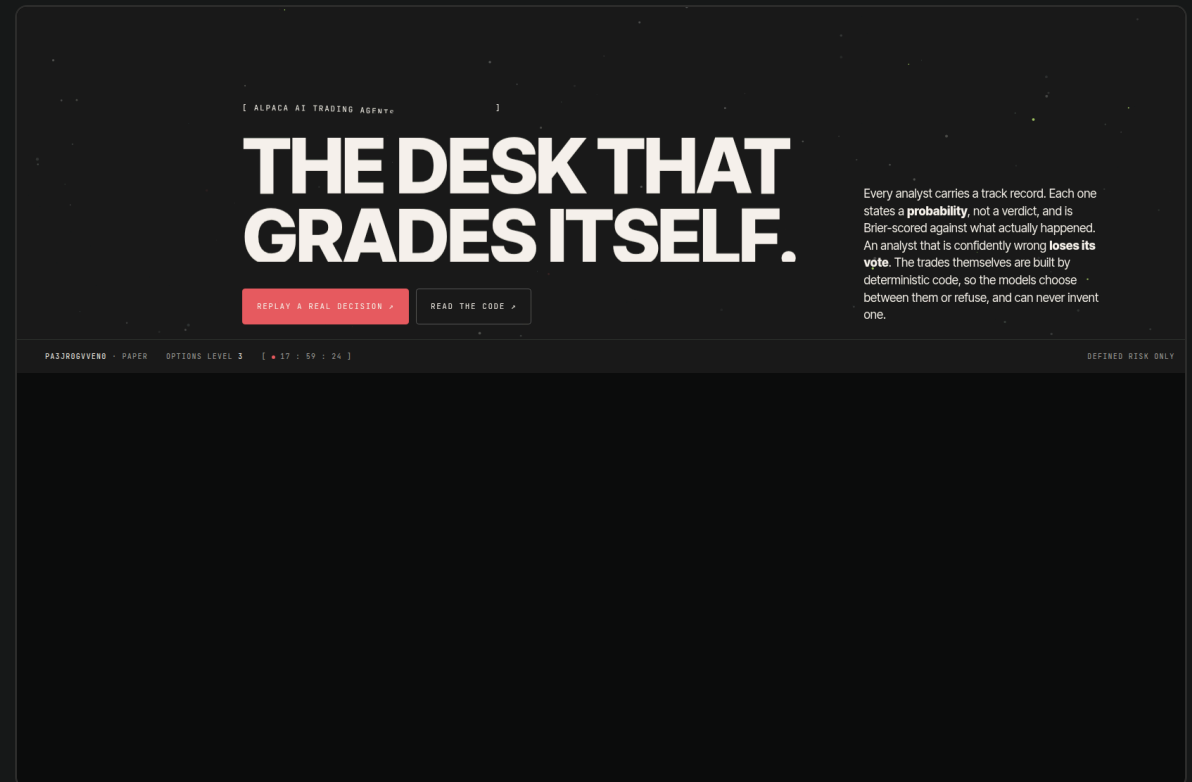


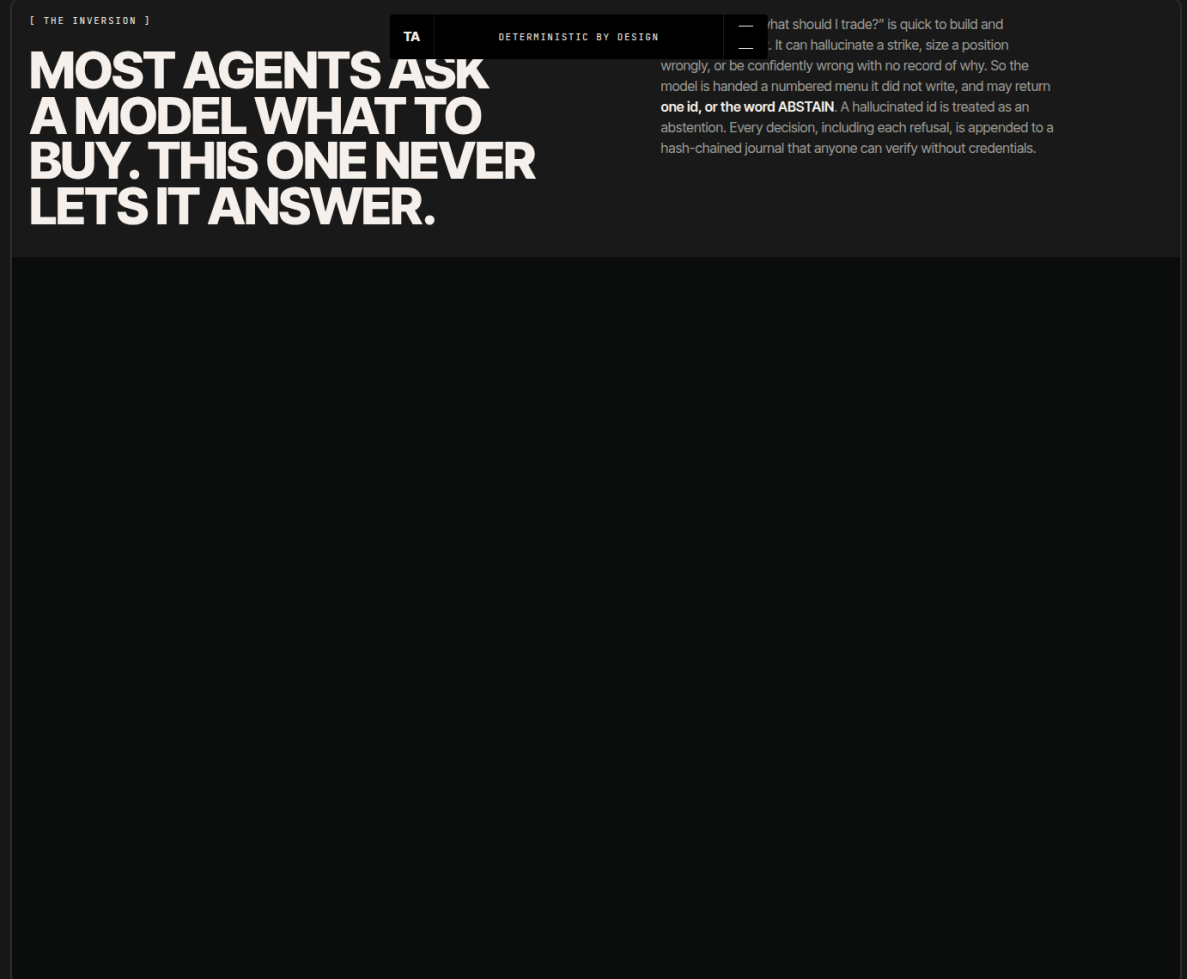
TRADING ALPACA: THE OPTIONS DESK THAT GRADES ITSELF

- Deterministic code builds every trade. A language model may only pick one by id, or refuse.
- Two independent vetoes · a fail-closed risk guard · a hash-chained journal · analysts that are Brier-scored and lose their vote when miscalibrated.
- Live site: <https://trading-alpaca-judge.vercel.app>
- Judge desk (no credentials needed): <https://trading-alpaca-judge.vercel.app/judge>
- Code: <https://github.com/Prashant-thakur77/trading-alpaca>

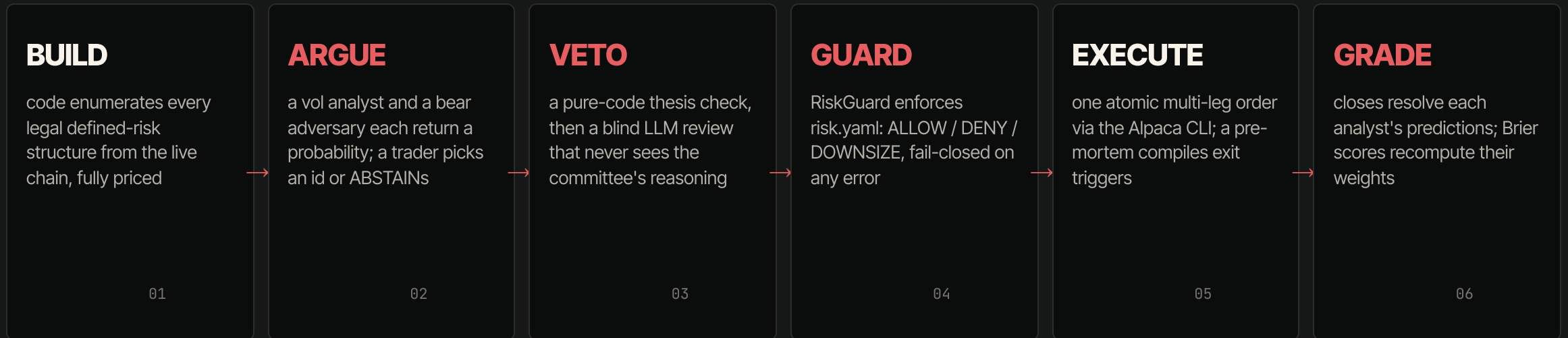


MOST AGENTS ASK A MODEL WHAT TO BUY. THIS ONE NEVER LETS IT ANSWER.

- Asking an LLM "what should I trade?" is fast to build and impossible to trust.
- It can hallucinate a strike, size a position wrongly, or be confidently wrong with no record of why.
- So this desk hands the model a numbered menu it did not write. It may return one id, or the word ABSTAIN.
- A hallucinated id is treated as an abstention. There is no code path by which a model can invent a strike, change a quantity, or move a price.



SIX STAGES. FOUR OF THEM CAN SAY NO.



Every stage appends to a hash-chained journal. Every refusal is recorded with its reason.

EVERY CANDIDATE IS FULLY SPECIFIED BEFORE ANY MODEL IS CALLED.

- Python reads the live SPY option chain and constructs every legal defined-risk structure: bull put, bear call, bull call and bear put spreads, iron condors, long straddles and butterflies.
- Each has strikes, legs, quantity, limit price, max loss and breakeven fixed by code.
- Liquidity gates first: open interest ≥ 100 , quote width $\leq 10\%$ of mid, 7–45 days to expiry.
- A live cycle reads ~3,700 contracts, ~1,100 pass liquidity, ~600–1,800 structures are built, 12 are shown to the committee.
- There is no function in the codebase that produces a naked short option.

01 **SNAPSHOT** What the desk saw

REALIZED VOLATILITY

9.52%

AT-THE-MONEY IMPLIED

11.78%

+2.27pp (implied is ABOVE realized — options are rich relative to actual movement, which favours SELLING premium, not buying it)

ID	STRUCTURE	DTE	CREDIT	MAX LOSS	BREAKEVEN
c1	bear_call_spread	27	\$2.76	\$224	772.76
c2	bear_call_spread	27	\$2.04	\$296	779.04
c3	bear_call_spread	32	\$2.12	\$289	779.12
c4	bear_call_spread	32	\$0.00	\$500	824.00
c5	bear_call_spread	34	\$2.83	\$217	772.83
c6	bear_call_spread	34	\$0.49	\$451	801.49
c7	bull_put_spread	20	\$1.97	\$303	767.03
c8	bull_put_spread	27	\$1.93	\$308	767.08
c9	bull_put_spread	32	\$0.00	\$500	520.00
c10	bull_put_spread	32	\$-0.03	\$503	555.03
c11	bull_put_spread	32	\$1.34	\$366	757.66
c12	bull_put_spread	34	\$1.30	\$370	755.70

Every one of these was built by deterministic code before any model saw anything. The committee may pick one by id or abstain. It cannot invent a strike, change a quantity, or move a price, because no code path allows it.

PROBABILITIES, NOT VERDICTS. AN ADVERSARY ARGUES AGAINST EVERY TRADE.

- vol_analyst judges implied vs realized volatility; bear_adversary's only job is to find what breaks the trade. Each returns a probability, and an abstaining analyst is excluded from both numerator and denominator.
- The trader picks one id or ABSTAINs. In this live cycle it moved from c1 to c2 because the adversary flagged c1's at-the-money short strike — the objection changed the trade, and is journalled either way.

```
=====
MODE: LIVE – orders WILL be sent to the broker this run.
=====
managing 1 open trade(s) before looking for a new one
  open   SPY bear_call_spread – no exit trigger fired (spot 761.72, 31 DTE, unrealized $+27.00)
SPY spot $761.72 – 1852 candidate(s)
ATM IV (chain-derived, ~30 DTE): 12.75%
mode: LLM COMMITTEE – vol_analyst + bear_adversary -> trader -> thesis veto + blind veto
committee:
  vol_analyst      p=0.79 – IV is 5.39pp rich to realized—a substantial signal to sell premium. Liquid credit spreads
(c8 bull put 29 DTE, c1 bear call 17 DTE, c2 bear call 45 DTE) capture this skew with solid breakeven distances and OI;
long-vol structures (c12 straddle, c10/c11 butterflies) buy expensive premium and fight the vol view.
  bear_adversary   p=0.50 – Despite the +5.39pp IV-RV edge favoring short premium, the extreme gap indicates realized
volatility may spike toward implied levels (12.75% vs 7.37%), pushing expected 17-day moves from $14.55 to $25+, which
breaches the max-loss strike at 767. The 762 short call is at-the-money with high assignment risk, and gap/earnings risk
over 17 DTE inadequately compensated by the $301 max profit versus $199 max loss—this is essentially a bet that realized
vol *stays* depressed, not a robust edge.
```

TWO REVIEWERS WITH DECORRELATED FAILURE MODES.

- Two calls to the same model on the same context agree with each other and prove nothing.
- Gate 1 is pure code: do the position's Greeks match the structure's own directional thesis? Unmeasurable Greeks fail closed.
- Gate 2 is an LLM deliberately starved of the committee's reasoning, shown only the trade's own facts.
- Live on Thursday: a candidate passed the thesis check and was vetoed by the blind review — "requires conviction SPY won't rally 1.4% over 44 days." The desk abstained.

ALLOW

DOWNSIZE

DENY

FAIL-CLOSED

No order was sent.

UNDERLYING n/a	SPOT n/a	ATM IV VS REALIZED n/a	CANDIDATES BUILT 0 shown	COMMITTEE CHOICE ABSTAIN
CONTRACTS 0				

WHY THIS IS NOT FULLY GREEN

- Upstream data could not be fetched. The desk reports an outage rather than a market judgement, and sends nothing.

RISK.YAML IS THE SINGLE SOURCE OF TRUTH. THE GUARD FAILS CLOSED.

LIMIT	VALUE
Max loss per position	\$1,000 (1% of equity)
Concurrent positions	3
New trades per underlying per day	1
Net delta / net vega	$ \Delta \leq 30 \cdot \nu \leq 200$
Days to expiry	7 – 45
Liquidity	$OI \geq 100 \cdot \text{spread} \leq 10\% \text{ of mid}$
Daily loss halt	2% of equity
After 3 consecutive losers	size halved
Any error, missing data, or exception	DENY

Verified by 933 tests, including mutation tests that prove a guard exception denies rather than trades. A kill-switch file halts everything within a minute.

IT MANAGES THE BOOK BEFORE IT ADDS TO IT.

- Orders are one atomic multi-leg limit through the official Alpaca CLI. Live fills came in at better than their limits (\$2.35 vs \$1.94, \$1.48 vs \$1.22).
- Before the trade is sent, a pre-mortem asks the model how it fails, then compiles those failure modes into deterministic exit triggers.
- Three exits apply to every position regardless of any model: 50% of credit captured, max loss, and a forced close at 3 DTE.
- Thursday: the breakeven trigger set at entry fired at 772.42 and the desk closed the spread itself at -\$70.50, before the loss could grow toward its \$352 max.

Thursday 3 Sep – the exit monitor fires

```
MODE: LIVE – orders WILL be sent to the broker this run.
=====
managing 3 open trade(s) before looking for a new one
  open  SPY bear_call_spread – no exit trigger fired (spot 772.94, 29 DTE, unrealized $-59.00)
  CLOSED SPY bear_call_spread ($-70.50) – underlying_beyond at 772.42 fired: Underlying breaks the breakeven threshold,
pushing the spread into losses at expiration
  open  SPY bear_call_spread – no exit trigger fired (spot 772.94, 43 DTE, unrealized $-16.00)
SPY spot $772.94 – 1848 candidate(s)
ATM IV (chain-derived, ~30 DTE): 12.█
```


AN ANALYST THAT IS CONFIDENTLY WRONG LOSES ITS VOTE.

ANALYST	RESOLVED	BRIER	WEIGHT	INTERPRETATION
vol_analyst	12	0.162	1.18	beats the 0.25 always-0.5 baseline — upweighted
bear_adversary	12	0.298	0.90	worse than baseline — down-weighted

Closes resolve the probabilities each analyst gave; Brier scores recompute weights every cycle and feed the next aggregate. Twelve resolved trades is a small sample and is stated as one — the loop is the point, not the score.

OVER 43 REPLAYED DECISIONS, IT REFUSED 72% OF THE TIME.

- The real committee replayed over 43 post-knowledge-cutoff windows: 31 refusals, 12 executable trades.
- Every refusal gave a reason. The reasons are what matter: after the structure menu was fixed, zero refusals cited a missing structure; seven of eight were the two reviewers failing to agree.
- Replayed at \$0.00: every LLM call is content-addressed and cached with its prompt, raw response and parse.
- This is not timidity. A desk that cannot say no cannot be trusted when it says yes.

[MEASURED OVER 43 REPLAYED DECISIONS]

IT REFUSED 72% OF THE TIME.

That is not timidity, and it is not a failure rate. Replaying the real committee over 43 post-cutoff windows, 31 ended in a refusal. Every one of them gave a reason, and the reasons are recorded in the journal whether or not they prevailed. Two examples, verbatim:

A REFUSAL THAT WAS A DEFECT

"Every available candidate is a short-premium credit structure, but implied vol sits below realized, a regime that argues for buying premium rather than selling it."

23 of 31 refusals said a version of this. The desk was right, and the menu was broken: the only long-premium structure it could build cost \$2,278 against a \$1,000 cap, so it was dropped before the committee ever saw it. Fixed.

A REFUSAL THAT WAS THE PRODUCT

"vol_analyst favors a bullish long-premium structure (cheap calls, c7 to c9) while bear_adversary flags material downside and gap risk. The two views don't converge on a single direction, so per veto policy that disagreement forces ABSTAIN."

After the fix, zero of eight refusals cite a missing structure. Seven of eight are the two reviewers failing to agree. That is the system working.

The rate did not fall when we fixed the menu: 8 of 10 before, 8 of 10 after, on matched dates. We are reporting that as measured. What changed is the reason, and the reason is the part that matters.

THE VOLATILITY SMILE — AND ITS NOISE FLOOR.

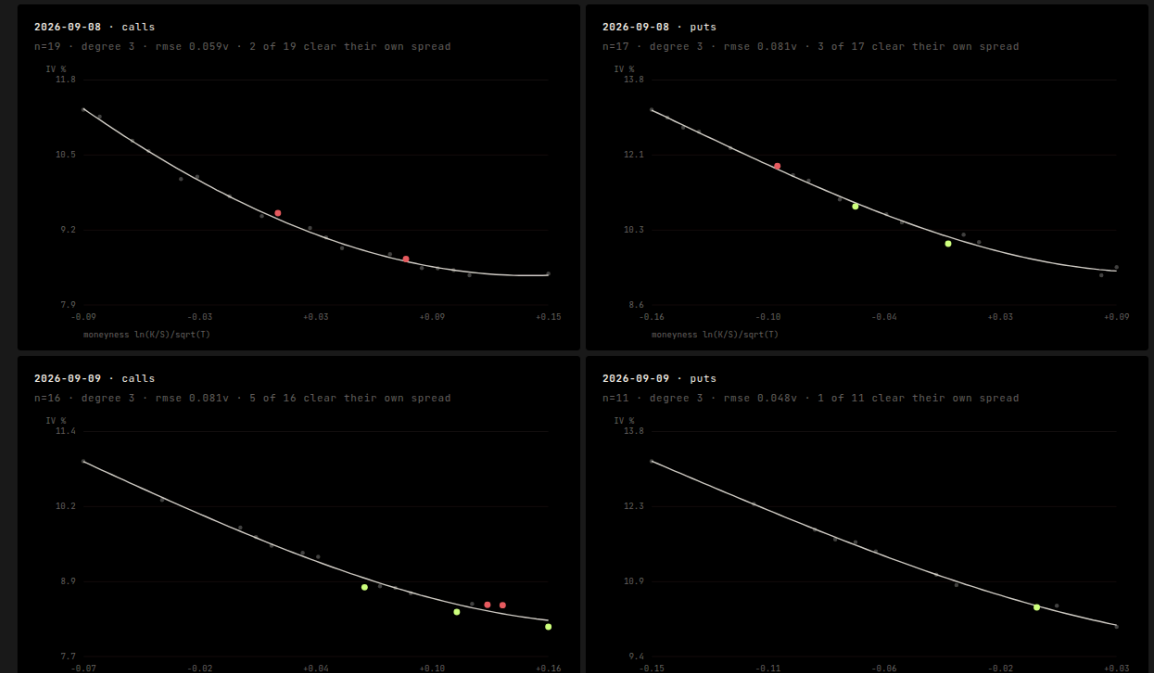
- A cubic fitted per expiry on $\ln(K/S)/\sqrt{T}$ across the liquid strikes only; 14 of 17 liquid groups fit.
- A strike is coloured only if its distance from the curve exceeds its own bid-ask spread. Everything grey scores exactly zero.
- We measured it, and deliberately did not let it pick strikes: the median deviation is worth about \$3.94 per contract against a \$4.00 quoted spread.
- Measurement is not edge, and the desk knows the difference.

[SPY · SPOT 749.35 · 14 OF 17 LIQUID GROUPS FITTED]

THE SMILE, AND ITS NOISE FLOOR.

Implied volatility per strike against moneyness, with a cubic fitted across the liquid strikes only. A dot is coloured **only if its distance from the curve exceeds that strike's own bid-ask noise**. Everything grey scores exactly zero and is ignored.

● rich · IV above the curve ● cheap · IV below the curve ● inside its own spread · scores zero



WHAT ACTUALLY HAPPENED.

Account	PA3JR0GVVEN0 — new, dedicated, paper, created 2026-08-29
Opening equity	\$100,000.00, flat until the window opened
Closing equity (Thu 3 Sep EOD)	\$99,827.64 (−0.17%)
Fills	3, all inside the window, each at better than its limit
Closes	1, automated, on its own breakeven trigger (−\$70.50)
Guard refusals	per-day cap, position cap, no-stacking
Positions corrupted	0
Every trade	a defined-risk bear call spread, 1 contract, inside its \$1,000 cap

Judged on total equity, this is flat. Three bear call spreads were one directional view expressed three times, and SPY rallied. We state that rather than hide it.

[FOUND BY RUNNING, NOT BY READING]

FOUR BUGS, ALL WHILE THE TEST SUITE WAS GREEN.

DAY	BUG	WHAT IT COST	FIX
Mon	Skill leak	the host's plugin told analysts to look for skills first → silent abstain	--setting-sources "" isolates the model call
Mon	Unfillable limit	mid-priced order rested 55 min, blocked every later cycle	20% marketable concession at the one live pricing site
Tue	Trailing comma	a valid, reasoned trade decision discarded over one character	string-aware JSON repair, fail-closed otherwise
Thu	Leg collision	new spread sold a contract already held long → broker rejected	filter candidates against current holdings

Each was root-caused from the journal and prompt cache, fixed with a regression test, and shipped mid-session. That is the argument for the journal.

STATED LIMITS, NOT OMITTED ONES.

- One closed trade and two open ones prove nothing statistically, and we do not imply otherwise.
- Paper fills are simulated against live quotes. No counterparty took the other side.
- The desk used ~25% of its own risk budget: candidates are built at 1 contract and the guard only ever downsizes. Known, documented, unchanged during scoring.
- Walk-forward backtests (13 windows, 90-day fit / 30-day test) are labelled as backtests; AAPL loses (PF 0.83) and the report says so.
- The previous author's hardcoded performance claims were deleted, not adapted.

[TRY IT YOURSELF]

EVERYTHING HERE IS VERIFIABLE WITHOUT CREDENTIALS.

Live site

<https://trading-alpaca-judge.vercel.app>

Judge desk — replay four real decisions, two of them refusals

<https://trading-alpaca-judge.vercel.app/judge>

Volatility smile

<https://trading-alpaca-judge.vercel.app/smile>

Source, tests, journal, and every prompt

<https://github.com/Prashant-thakur77/trading-alpaca>

Demo video (4:22)

<https://github.com/Prashant-thakur77/trading-alpaca/raw/main/docs/press/demo.mp4>

```
git clone https://github.com/Prashant-thakur77/trading-alpaca
cd trading-alpaca && make install
make test           # 933 tests, offline
make judge          # 4 judge scenarios, no keys
make verify-journal # hash chain intact
```

Compliance: orders route through the official Alpaca CLI; market data through the official Alpaca MCP server with read-only toolsets (no trading toolset is exposed, so an LLM cannot place an order through it). alpaca-py is used for analysis only.